SYSTEM SPECIFICATION: Sharma Absolute Boundary Summation (ABS) Model Project

Act as an expert Software Architect and Physics Researcher. Your task is to generate a complete GitHub project repository for the **Sharma Absolute Boundary Summation (ABS) Model**, an original multiplicative noise heuristic developed by Nitish Sharma.

1. MATHEMATICAL CONTEXT

The model produces a noise factor η:

$$\eta = \frac{n(|\psi_{min}| + |\psi_{max}|)}{n(|\psi_{min}| + |\psi_{max}|) + \sum_{i=1}^{n} r_i}$$

Where $r \in [\psi_{min}, \psi_{max}]$. The final signal is: $V_{real} = V_{ideal} \cdot [1 + \alpha(\eta - 1)]$.

2. PROJECT STRUCTURE REQUIREMENTS

Generate the following files with complete, production-ready code:

A. Core Library (C/Embedded) - `src/sharma_abs.h`

Create a header-only or source implementation optimized for microcontrollers. It must use basic arithmetic and avoid expensive math libraries. Include functions for:

- `compute_eta(min, max, n)`
- `apply_sharma_noise(ideal_val, eta, alpha)`

B. Core Library (Python) - `src/sharma_abs.py`

A class-based implementation of the model. Include methods to generate arrays of noisy data for any given input function (sin, linear, quadratic).

C. Web Application - `public/index.html`

A single-file interactive dashboard (HTML5/Tailwind/Chart.js).

- **Features**: Real-time graph comparing "Deterministic Ideal" vs "Sharma ABS Observation".
- **Interactivity**: Sliders for ψ_{min}, ψ_{max}, n (Complexity), and α (Influence).
- **Presets**: Dropdown for different physics models (Free Fall, Damped Oscillation, Harmonic Motion, Financial Shocks).
- **Visuals**: A "Math Box" showing live calculation values.

D. System Application (CLI Tool) - `tools/abs_gen.py`

A Python CLI utility that:

1. Takes a CSV of deterministic data.
2. Applies the Sharma ABS Model based on user flags (`--alpha` , `--complexity`).
3. Exports the noisy "real-world" dataset to a new CSV and generates a PNG plot.

3. RESEARCH & DOCUMENTATION

- **README.md**: Use the provided branding "Sharma ABS Model". Explain the "Stabilizing Constant" in the numerator and the "Magic 11" quantization effect (bottleneck at $n = 12$). Include the official viXra citation (ID: 17827727).
- **LICENSE**: Standard MIT License.

4. KEY CONSTRAINTS

- **Branding**: Always use the term "Sharma ABS Model".
- **Stability**: The code must handle the "Magic 11" effect gracefully and ensure no division by zero.
- **Visuals**: Use professional color palettes (Indigo, Emerald, Rose).